



Course Modification: CIT 82 - Introduction to AI and Web Development

Subject CIT

Discipline(s):

Computer Information Systems (Computer network installation, microcomputer technology, computer applications)

Formerly:

Cross Listed Courses:

Term Effective: 2026 Spring Semester

Course Control Number (CB00): CCC000457233

CB03: Top Code: 0709.00 - World Wide Web Administration*

CB09: Sam Code: D - Possibly Occupational

CC Approval: 10/17/2022

Board of Trustees: 12/13/2022

Units/Hours

Total Units: 3.00

Lecture Units: 3.00

Lab Units: 0.00

Activity Units: 0.00

Number of Weeks 18

Lecture Hours Per Week: 3.00

Lab Hours Per Week: 1.00

Activity Hours: 0.00

Total In Class Contact Hours: 72.00

Total Out of Class Contact Hours: 108.00

Total Student Learning Hours: 180.00

Number of Repeats for Credit 0

Grading Method Letter Grade

Open Entry/Open Exit No

Requisites

Prerequisite: None
Corequisite: None
Advisory: CIT 15 CIT 48A
Anti Requisite: None

Course Description

This course surveys contemporary web technologies and the modern digital ecosystem. Students will explore foundational concepts while learning to leverage AI tools for planning, analysis, and development workflows. Topics include: web servers, scripting, protocols, E-Commerce, information architecture, modern traffic analysis, search engine optimization, and website usability. (A, CSU)

Transfer/General Education

Requested Credit Classification: D - Credit - Degree Applicable

Transfer Status (CB05): B - Transferable to CSU only.

CSU Transfer Course

- CSU Transfer

CSU Transfer Course

- CSU Transfer Course

C-ID:

Objectives

1. Describe contemporary web technologies and AI tools used in website creation and delivery.
2. Identify the parties and components involved in e-commerce systems.
3. Describe current theories on search engine optimization and the impact of AI on search discovery.
4. Create a website requirement business planning document utilizing AI-assisted analysis.
5. Create a simple website utilizing modern frameworks, web standards, and AI prompting to assist in implementation.
6. Generate digital assets and text content for website creation utilizing AI tools.

Methods of Instruction

Multiple Methods of Instruction that may be used to provide multiple means of representation of course content in alignment with Universal Design for Learning principles.

- Demonstration/Return Demonstration
- Lecture/Discussion
- Laboratory

Assignments

Multiple representative assignments that may be used to provide multiple means of engagement and expression in alignment with Universal Design for Learning principles.

Required Reading

Foundational Web Documentation:

Selected modules from the Mozilla Developer Network (MDN) Web Docs focusing on HTML5 semantics, CSS core concepts, and introductions to modern layout frameworks.

W3C Web Accessibility Initiative (WAI) guidelines to understand inclusive design and universal access on the web.

AI Prompting and Digital Ecosystem Resources:

Industry-standard open-source documentation on effective prompting techniques for technical code generation and text optimization (e.g., LearnPrompting.org or platform-specific guidebooks).

Current tech industry articles and case studies exploring how Large Language Models (LLMs) and AI-driven search engines alter traditional Search Engine Optimization (SEO) and user search behavior.

Required Writing

Website Usability and Functionality Audit:

Students will analyze an existing commercial or non-profit website against modern web standards, accessibility requirements (WCAG), and user interface guidelines. They will synthesize their findings into a structured evaluative report or annotated digital presentation proposing specific improvements.

AI Prompting and Iteration Logs:

While utilizing AI prompting to assist in implementing a website layout or generating digital text/media assets, students will maintain a reflective documentation log. This log requires writing out the initial prompt, analyzing why the AI output did or did not meet technical standards, and detailing the revised prompts used to achieve the accurate final result.

E-Commerce Business Requirements Document:

Students will draft a website requirement business planning document for a proposed web project. This written proposal will outline the target audience, information architecture, required e-commerce protocols (HTTP vs. HTTPS), merchant options, and an updated search engine optimization (SEO) strategy that accounts for modern AI search discovery.

Examples of Out of Class Reading and Writing

Structured Technical Note-Taking (Markdown):

Writing: Students will use Markdown syntax to document and structure their weekly notes on core web technologies (such as domains, HTTP/HTTPS protocols, and web standards). These documents will be pushed to a repository or shared digitally to practice modern content-authoring formats.

AI Prompting Critique and Refinement:

Reading: Students will read industry blogs or case studies regarding common pitfalls in AI-generated code and web copy (e.g., semantic errors, outdated framework syntax, or biased outputs).

Writing: Given a poorly constructed prompt and an inaccurate AI-generated code snippet, students will write an analysis explaining why the prompt failed and draft a refined, structured prompt to correct the output.

E-Commerce Case Study Analysis:

Reading: Students will read provided documentation and real-world case studies detailing different e-

commerce implementations (e.g., hosted shopping carts vs. custom databases, payment gateway security protocols).

Writing: Students will complete a short composition comparing two different business models, explaining the security measures (HTTPS) and business policies required for each to protect user data.

Other

Critical Thinking:

Choosing web server platforms and selecting protocols needs for a specific business website requirement(s).

Choosing the e-commerce software based on business needs.

Evaluating and verifying AI-generated outputs for accuracy, bias, and adherence to web standards.

Methods of Evaluation

Multiple Methods of Evaluation that may be used to provide multiple means of engagement and expression in alignment with Universal Design for Learning principles.

Written Assignments

Substantial writing assignments are required (25% or more of grade).

Informal Essay/Short Composition

AI Collaboration and Prompt Engineering Reflection Logs: Students will submit short, informal written reflections detailing their experience using AI prompting to assist in implementing website elements. The composition must analyze their prompt structure, evaluate the accuracy and web-standard compliance of the AI's output, and explain the iterative steps taken to correct any errors.

Traditional vs. AI-Driven SEO Comparative: Students will write a short composition analyzing how the rise of Large Language Models and AI search overviews alters traditional, keyword-focused Search Engine Optimization (SEO). They will discuss how modern web developers must adapt their data architecture and content strategies to remain discoverable in an AI-driven digital ecosystem.

Noncomputational

Information Architecture and Sitemap Blueprinting: Given a specific business scenario or project requirement, students will logically organize complex data into a cohesive information architecture. They will determine static versus dynamic page requirements and map out a visual flowchart or structural hierarchy (sitemap) that optimizes user navigation and web usability.

AI Output Verification and Bug Identification: Students will be presented with an AI-generated code snippet or website layout structure that contains logical errors, deprecated tags, or accessibility violations. Students must systematically analyze the output, diagnose the non-computational flaws (such as broken semantic structures or poor contrast choices), and explain how to correct them to align with current web standards.

E-Commerce Pipeline and Technology Selection: Students will analyze the needs of a hypothetical business to design an e-commerce workflow. They must determine the necessary components—including shopping cart software compatibility, payment gateway requirements, merchant account types, and specific security protocols—justifying their architectural choices based on operational constraints.

Hands-on

Framework Implementation and Content Creation: Students will build a responsive, multi-page website utilizing modern design frameworks and web standards. They will actively use structured AI prompting techniques to help generate semantic content layouts, responsive components, and accessible navigation menus, demonstrating the ability to translate conceptual prompts into clean,

functional web projects. Version Control and Web Server Deployment: Students will initialize local Git repositories for their web projects, track code revisions, and write descriptive commit messages. They will successfully push their localized code to a cloud-based hosting environment or web server (such as GitHub Pages or a similar platform), demonstrating mastery of modern digital distribution workflows. Digital Asset and Text Generation Integration: Students will construct a website layout that requires custom media and copy. They will demonstrate hands-on proficiency with generative AI tools to produce targeted digital text assets and optimized web imagery, embedding these assets into the framework structure while maintaining proper file-pathing and web accessibility compliance (such as alt-text headers).

Texts, Other Readings, and Materials:

Other:

Open Educational Resources All instructional materials, including current online industry resources, technical framework documentation, and instructor-created tutorials, will be provided to students digitally at no cost. Materials are continuously curated by the instructor to reflect active industry standards published within the last 5 years.

Course Content

Lecture Content:

Lecture Content Outline

- **I. Introduction to Web Technologies**
 - A. Markup Languages (HTML5 structure and semantic tags)
 - B. Domain Name System (DNS) and web hosting concepts
 - C. Hypertext Transfer Protocol (HTTP) and client-server architecture
 - D. Web Standards, World Wide Web Consortium (W3C), and browser compatibility
 - E. Open-source ecosystems and development tools
- **II. E-Commerce Fundamentals**
 - A. Product catalog structures and digital inventories
 - B. Shopping cart software and checkout workflows
 - C. Secure Hypertext Transfer Protocol (HTTPS) and SSL/TLS encryption
 - D. Merchant accounts, payment gateways, and transaction processing
- **III. Information Architecture and AI-Assisted Planning**
 - A. Website layout planning and sitemaps
 - B. AI-assisted project planning and structural brainstorming
 - C. Foundational AI prompt engineering techniques for content organization
- **IV. Web Traffic and Modern Analytics**
 - A. Server logs, request-response tracking, and data collection
 - B. Modern analytics platforms and cloud-based tracking dashboard
- **V. Search Engine Optimization (SEO) and the Evolution of Discovery**
 - A. Traditional SEO mechanics: keywords, metadata, and header structures
 - B. Content optimization strategies for readability and indexing
 - C. Search engine web crawlers, spiders, and indexing behavior
 - D. Pay-per-click (PPC) and search engine marketing (SEM) models

- E. AI Search Discovery: optimizing for Large Language Models (LLMs) and AI search overviews
- **VI. Website Usability, Accessibility, and Content Generation**
 - A. User Interface (UI) and User Experience (UX) design principles
 - B. Web accessibility standards (WCAG) and universal inclusive design
 - C. Generative AI tools for digital asset generation (custom imagery, layouts, and copy)
 - E. Critical evaluation and verification of AI-generated assets for accuracy and compliance

Lab Content:

Lab Supports Lecture

- **I. Introduction to Web Technologies**
 - A. Markup Languages (HTML5 structure and semantic tags)
 - B. Domain Name System (DNS) and web hosting concepts
 - C. Hypertext Transfer Protocol (HTTP) and client-server architecture
 - D. Web Standards, World Wide Web Consortium (W3C), and browser compatibility
 - E. Open-source ecosystems and development tools
- **II. E-Commerce Fundamentals**
 - A. Product catalog structures and digital inventories
 - B. Shopping cart software and checkout workflows
 - C. Secure Hypertext Transfer Protocol (HTTPS) and SSL/TLS encryption
 - D. Merchant accounts, payment gateways, and transaction processing
- **III. Information Architecture and AI-Assisted Planning**
 - A. Website layout planning and sitemaps
 - B. AI-assisted project planning and structural brainstorming
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 - A. Server logs, request-response tracking, and data collection
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- **VI. Website Usability, Accessibility, and Content Generation**
 - A. User Interface (UI) and User Experience (UX) design principles
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 - C. Generative AI tools for digital asset generation (custom imagery, layouts, and copy)
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Student Learning Outcomes

1. Demonstrate an understanding of contemporary web technologies and AI prompting techniques in relation to authoring, distributing, and browsing.

Diversity, Equity, and Inclusion

Description of approaches faculty may use to accommodate and engage diverse student bodies, advance equitable student outcomes, and promote the inclusion of all students (Title 5 § 55001).

Description of Approaches

Faculty promote diversity, equity, and inclusion in CIT 82 through student-centered instructional practices that support learners with varied technical backgrounds, preparation levels, and educational goals. Course materials and evaluations are structured to eliminate historical barriers to technology entry through the deployment of early diagnostic success audits and supportive bridge labs. By integrating structured AI prompting techniques, the curriculum democratizes access to digital creation, allowing students without prior syntax or programming exposure to successfully brainstorm, organize, and author web content. Course topics are consistently connected to diverse, real-world professional contexts, and zero-cost digital resources ensure financial barriers do not impede student equity, fostering an inclusive environment for all learners entering the modern digital workforce.